Kharif 2026 drought assessment: Maharashtra

Pik-drishti AI · data to 2026-10-08 · following the Manual for Drought Management (2016)

Item	Value
Units assessed	365 talukas
Trigger-1 met	287
Severe / moderate (Trigger-2)	0 / 7
Rainfall window	2026-06-01 to 2026-09-30
Crop-condition window	2026-09-22 to 2026-10-08

Method

Trigger 1: Rainfall 20% or more below the 1991-2020 normal with a dry spell of 3+ weeks (each week under 50% of normal), or 60% or more below normal (Manual Table 3.10).

Remote sensing: NDVI and NDWI over the latest 24 days vs the unit's normal-rainfall years (2017-2025); -20 to -30% moderate, below -30% severe; worst of the two (Manual 3.2.2).

Soil moisture: Moisture Adequacy Index = AET / PET from MODIS, expressed as % of the unit's normal-year value, then Manual Table 3.5 (<=25 severe, 26-50 moderate, 51-75 mild).

Agriculture: Share of cropland whose August-to-date peak NDVI exceeds 0.6, vs normal years; same deviation classes as remote sensing.

Category: Severe if all three impact indicators are severe; moderate if two of three are moderate or severe; otherwise normal (Manual, Trigger-2). Field verification (Step 3) decides the final category.

Adaptations: MODIS PET is a land-surface potential, not the crop water need used in the manual's weekly water balance, so MAI is compared with the unit's own normal years. Cropped area is a satellite estimate, not Agriculture Department sowing returns.

Results

Unit	District	Drought category	Trigger-1	Rain (mm)	Normal (mm)	Rain deviation %	Longest dry spell (weeks)	Dry spell from	NDVI deviation %	NDWI deviation %	VCI
Mohol	Solapur	moderate	True	281.0	451.0	-37.7	4	2026-08-24	-9.8	-21.2	58.4
Sangola	Solapur	moderate	True	214.0	342.0	-37.3	3	2026-06-29	-16.6	-34.6	50.6
Mangalvedhe	Solapur	moderate	True	239.0	376.0	-36.3	4	2026-08-24	-18.4	-36.1	41.3
Atpadi	Sangali	moderate	True	304.0	465.0	-34.6	3	2026-08-24	-7.7	-22.9	69.2
Solapur North	Solapur	moderate	True	374.0	540.0	-30.7	3	2026-08-24	-10.8	-21.7	52.6
Jath	Sangali	moderate	True	274.0	387.0	-29.3	4	2026-08-24	-14.6	-33.7	54.3
Pune City	Pune	moderate	True	633.0	692.0	-8.6	4	2026-08-24	-8.2	-23.6	38.6
Nevasa	Ahilyanagar	normal	True	240.0	490.0	-51.1	7	2026-08-03	0.6	12.5	77.3
Gangapur	Chhatrapati Sambhajinagar	normal	True	253.0	512.0	-50.6	8	2026-08-03	-9.8	-11.1	57.8
Chauka	Chhatrapati Sambhajinagar	normal	True	332.0	665.0	-50.2	8	2026-08-03	2.0	13.3	66.0

Unit	District	Drought category	Trigger-1	Rain (mm)	Normal (mm)	Rain deviation %	Longest dry spell (weeks)	Dry spell from	NDVI deviation %	NDWI deviation %	VCI
Fulambari	Chhatrapati Sambhajinagar	normal	True	323.0	641.0	-49.6	7	2026-08-03	7.5	28.9	83.1
Aurangabad	Chhatrapati Sambhajinagar	normal	True	328.0	650.0	-49.5	8	2026-08-03	-6.4	-7.8	44.2
Paithan	Chhatrapati Sambhajinagar	normal	True	296.0	587.0	-49.5	6	2026-08-17	-7.6	-8.3	60.5
Rahata	Ahilyanagar	normal	True	265.0	524.0	-49.4	7	2026-08-03	6.6	29.7	75.1
Khuldabad	Chhatrapati Sambhajinagar	normal	True	318.0	623.0	-48.9	8	2026-08-03	7.3	22.2	86.0
Shrirampur	Ahilyanagar	normal	True	267.0	522.0	-48.7	7	2026-08-03	2.0	15.9	73.9
Shevgaon	Ahilyanagar	normal	True	276.0	533.0	-48.3	6	2026-08-17	-11.3	-16.3	57.3
Kannad	Chhatrapati Sambhajinagar	normal	True	329.0	633.0	-48.0	8	2026-08-03	3.0	16.0	77.7
Part of Vaijapur	Chhatrapati Sambhajinagar	normal	True	296.0	568.0	-47.9	8	2026-08-03	-1.4	8.4	73.3
Vaijapur	Chhatrapati Sambhajinagar	normal	True	279.0	519.0	-46.3	8	2026-08-03	-8.8	-14.1	58.8
Deulgaon-Raja	Buldana	normal	True	378.0	701.0	-46.1	7	2026-08-03	8.4	36.7	87.6
Kopargaon	Ahilyanagar	normal	True	292.0	540.0	-46.0	7	2026-08-03	2.9	13.8	72.7
Badnapur	Jalna	normal	True	349.0	644.0	-45.8	4	2026-08-17	1.6	12.1	70.8
Part of Jafrabad	Jalna	normal	True	369.0	680.0	-45.8	4	2026-08-17	6.6	26.9	86.9
Sakri	Dhule	normal	True	422.0	774.0	-45.5	3	2026-08-17	-6.1	-9.2	62.3
Sindkhed-Raja	Buldana	normal	True	383.0	701.0	-45.4	7	2026-08-03	1.7	19.7	70.5
Walwa	Sangali	normal	True	432.0	787.0	-45.1	3	2026-08-31	5.2	13.4	83.3
Ambad	Jalna	normal	True	343.0	621.0	-44.7	5	2026-08-17	-2.6	-2.1	57.7
Shirur-Kasar	Beed	normal	True	308.0	555.0	-44.6	5	2026-08-17	-11.7	-22.7	46.1
Nandgaon	Nashik	normal	True	314.0	565.0	-44.5	8	2026-08-03	-9.9	-11.7	44.6
Baglan	Nashik	normal	True	419.0	752.0	-44.3	3	2026-08-17	3.6	9.8	77.6
Jalna	Jalna	normal	True	359.0	643.0	-44.1	6	2026-08-03	-5.9	-4.6	48.4
Pathardi	Ahilyanagar	normal	True	272.0	485.0	-44.0	5	2026-08-17	-10.6	-21.8	54.3
Palus	Sangali	normal	True	344.0	614.0	-43.9	3	2026-08-31	6.4	19.3	85.3
Parbhani	Parbhani	normal	True	445.0	790.0	-43.6	5	2026-08-10	5.2	36.7	77.9
Yevla	Nashik	normal	True	307.0	536.0	-42.8	7	2026-08-03	5.2	24.4	73.7
Pandharpur	Solapur	normal	True	221.0	385.0	-42.8	4	2026-08-24	-4.2	-2.0	66.3

Unit	District	Drought category	Trigger-1	Rain (mm)	Normal (mm)	Rain deviation %	Longest dry spell (weeks)	Dry spell from	NDVI deviation %	NDWI deviation %	VCI
Jafrabad	Jalna	normal	True	401.0	700.0	-42.7	4	2026-08-17	11.3	43.7	90.0
Manwat	Parbhani	normal	True	428.0	745.0	-42.6	6	2026-08-03	5.5	22.4	80.8
Rahuri	Ahilyanagar	normal	True	308.0	536.0	-42.5	7	2026-08-03	-0.1	12.6	71.4
Chikhaldara	Amravati	normal	True	577.0	1002.0	-42.5	5	2026-08-03	2.8	25.2	83.0
Malegaon	Nashik	normal	True	350.0	608.0	-42.5	3	2026-08-17	-9.2	-12.0	53.9
Malegaon	Washim	normal	True	438.0	762.0	-42.5	6	2026-08-03	-3.2	1.7	55.7
Barshitakli	Akola	normal	True	411.0	714.0	-42.4	6	2026-08-03	-12.4	-17.0	28.3
Chalisgaon	Jalgaon	normal	True	372.0	646.0	-42.4	6	2026-08-17	-8.4	-7.4	53.0
Lonar	Buldana	normal	True	419.0	724.0	-42.1	7	2026-08-03	4.2	37.9	73.1
Mehkar	Buldana	normal	True	427.0	737.0	-42.0	7	2026-08-03	5.4	53.3	72.1
Hatkalangale	Kolhapur	normal	True	390.0	669.0	-41.7	3	2026-08-31	6.4	22.8	84.9
Basmath	Hingoli	normal	True	509.0	871.0	-41.6	5	2026-08-10	-2.2	16.1	52.9
Bhokardan	Jalna	normal	True	393.0	674.0	-41.6	4	2026-08-17	5.2	24.1	79.1
Purna	Parbhani	normal	True	499.0	852.0	-41.4	5	2026-08-10	-2.2	18.7	49.7
Mantha	Jalna	normal	True	404.0	689.0	-41.3	6	2026-08-03	-2.2	9.4	59.8
Karvir	Kolhapur	normal	True	555.0	945.0	-41.3	3	2026-08-31	6.4	24.9	80.9
Patoda	Beed	normal	True	307.0	522.0	-41.2	7	2026-08-03	-7.2	-25.7	50.3
Sillod	Chhatrapati Sambhajinagar	normal	True	390.0	662.0	-41.0	6	2026-08-03	7.8	28.9	82.8
Jintur	Parbhani	normal	True	453.0	768.0	-41.0	6	2026-08-03	-1.0	18.8	58.4
Georai	Beed	normal	True	363.0	614.0	-40.8	4	2026-08-17	-11.8	-19.8	51.6
Ghansavangi	Jalna	normal	True	388.0	652.0	-40.6	5	2026-08-17	-10.3	-13.2	44.0
Patur	Akola	normal	True	425.0	715.0	-40.5	6	2026-08-03	-5.4	2.1	48.5
Akola	Akola	normal	True	398.0	665.0	-40.2	5	2026-08-03	-4.6	3.0	55.6

Data sources

Dataset	Provider	Used for	Licence / citation
CHIRPS v2.0 daily rainfall	UC Santa Barbara Climate Hazards Center	Seasonal rainfall deviation and dry spells (normal: 1991-2020)	Public domain (CC0). Funk et al. (2015), Scientific Data 2:150066
MODIS Terra surface reflectance 8-day, MOD09A1 v6.1	NASA LP DAAC	NDVI and NDWI deviation, VCI (2001-2025), cropped area	NASA, free and open. Vermote (2021), MOD09A1 v061, doi:10.5067/MODIS/MOD09A1.061

Dataset	Provider	Used for	Licence / citation
MODIS Terra evapotranspiration 8-day, MOD16A2 v6.1 (and gap-filled MOD16A2GF)	NASA LP DAAC	Moisture Adequacy Index (AET / PET)	NASA, free and open. Running et al. (2021), MOD16A2 v061, doi:10.5067/MODIS/MOD16A2.061
ESA WorldCover 2021 v200	European Space Agency	Cropland mask (class 40)	CC BY 4.0. Zanaga et al. (2022), doi:10.5281/zenodo.7254221
Manual for Drought Management (2016)	Ministry of Agriculture & Farmers Welfare, Government of India	Indicators, thresholds, Trigger-1 and Trigger-2 rules	Government of India.
Indian Village Boundaries, Maharashtra (Census 2001 villages)	DataMeet community	Village boundaries; talukas and districts are built by merging villages. Some taluka names and boundaries have changed since 2001.	ODbL 1.0.

A satellite screening to support Steps 1-2 of the manual. The final category is decided after field verification (Step 3) by the State.